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Youzhi Wang

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Education

Columbia University
MS Computer Science
GPA: 3.96

New York, NY
Dec 2024

relevant courses: Data Structures and Algorithms , DataBases

University of California, Santa Barbara
BS (double major) Applied Mathematics, Statistics and Data science
relevant courses: Cryptography, Automata and Formal Languages
Graduated with Honors and Major(Applied Math) distinction

Santa Barbara, CA
June 2023

Projects

NATURAL LANGUAGE PROCESSING PROJECT

Creating robust datasets for LLM evaluation (New York, NY) **May 2024 – August 2024**

- Coauthored a paper for EMNLP 2024 conference (top conference in NLP)
- Evaluated current SOTA LLMs reasoning capabilities and found that data leaks during training is prevalent
- Co-created a dataset that better evaluates an LLMs reasoning capability
- Generated 400 math questions and answers for evaluating LLMs using Python scripts
- Parameterized math question using Python to introduce randomness to better evaluate LLM's reasoning ability

DEEP LEARNING PROJECTS

Schizophrenia classification (New York, NY) **Jan 2024 – May 2024**

- Trained Neural Networks on 3D MRI brain images that achieved AROC of 0.87 using PyTorch
- Visualized regions of the brain that cause schizophrenia using Python and JupyterNotebooks
- Compared the performance of 2D vision transformers and 3D Convolutional Neural Networks using Python scripts and found that 3D CNN models still performs better for brain MRIs

Cancer cell detection and classification (New York, NY) **Jan 2024 – May 2024**

- Fine tuned vision transformer model to detect position of cancer cell in a image using PyTorch
- Wrote Python scripts that utilizes Amazon Web Services to train Neural Networks

Multiple Sclerosis classification using Brain MRI (New York, NY) **Sep 2023 – Jan 2024**

- Extracted Radiomics features from 100 brain MRIs using Python and the PyRadiomics package
- Segmented 100 brains from MRI images using FreeSurfer and Bash scripts
- Wrote Python scripts using PyTorch and MONAI for classification tasks
- Ran Python and Bash scripts on a cluster computer through SSH and Sun Grid Engine

Earthquake data collection and classification (Santa Barbara, CA) **Jul 2022 – Jun 2023**

- Received Applied math Major Distinction award from the UCSB for my undergraduate thesis
- Collected and cleaned large collections of seismic signals (over 100,000 waveforms) using Python and ObsPy
- Improved high-quality signal identification time by 5 times through restructuring the previous data processing pipeline using Python JupyterNotebooks
- Labeled 200+ high-quality Earthquake signals using Python scripts
- Implemented 5 different Machine Learning algorithms (KNN, LDA/QDA, SVMs and DCNN) for classifying high quality seismic wave signals in Python JupyterNotebooks

OTHER CODING PROJECTS

HTTP 1.0 Web server **Sep 2023 – Dec 2023**

- Wrote a HTTP 1.0 web server using C and socket APIs with image hosting and an interactive search module that uses a linked list to search through a given database.

Command Line Anagram Finder **Nov 2023 – Dec 2023**

- Created a command line Java application to find all the anagrams of a given word from a large text document using Binary Search Trees, AVL Trees and HashMaps

SKILLS

Technical tools: Python, PyTorch, sklearn, Java, Bash, SQL, AWS, Linux, Git, C/C++, Vim/Neovim, Tmux
Other languages: Mandarin(Bilingual)